

COMPUTER SCIENCE TRIPOS Part IB – 2022 – Paper 6

3 Complexity Theory (ad260)

- Space Complexity (a) If A and B are decision problems, we write $A \leq_L B$ to denote that A is reducible to B by means of a *logarithmic-space* reduction. Give a precise definition of such a reduction. [2 marks]

Answer: A is logarithmic-space reducible to B if there is a 3-tape Turing machine M with a read-only input tape, a write-only output tape (i.e. one with a one-directional head) and a work tape. Started with a string x on its input tape and empty work and output tape, M eventually halts with a string $M(x)$ on its output tape with the property that $M(x) \in B$ if, and only if, $x \in A$. Moreover, the total number of tape cells used on the work tape in the computation is $O(\log |x|)$.

- Space Complexity. Reductions. (b) For decision problems A , B and C , show that if $A \leq_L B$ and $B \leq_L C$, we have $A \leq_L C$. [5 marks]

Answer: Suppose we are given a machine M witnessing $A \leq_L B$ and a machine N witnessing $B \leq_L C$. Let p be a polynomial such that the length of $M(x)$ is bounded by $p(|x|)$ (such a polynomial exists by the space bound on M). We construct a machine M' witnessing $A \leq_L C$. The machine has an input tape with a string x and an output tape on which we aim to produce $N(M(x))$. The working memory consists of a counter ($\log p(|x|)$) bits for the head position of N on its input tape, a counter ($\log p(|x|)$) bits for the head position of M on its output tape, plus the work tapes of both M and N . This is a total of $O(\log |x|)$ tape cells.

M' works by simulating a run of N . Whenever the simulated N attempts to read from its input tape at position i , M' starts a new run of M on input x , discarding all output until the symbol at position i on the output is produced. This is then used to carry out the next step of the simulation of N .

- Reductions and Completeness. NP-completeness. coNP. (c) For each of the four complexity classes P , NP , NL and $co-NP$, give an example of a problem that is *complete* for the complexity class under logarithmic-space reductions. You do not need to prove the completeness. [4 marks]

Answer: The circuit value problem (CVP) is P -complete. Satisfiability of Boolean formulas (SAT) is NP -complete. Reachability (REACH) in directed graphs is NL -complete. Validity of Boolean formulas (VAL) is $co-NP$ -complete.

Many other examples are possible and any correct ones will be accepted.

- Reductions and Completeness. Reachability Method. Time and Space Complexity. (d) For each pair of problems A and B from your answers to part (c) above, state whether or not $A \leq_L B$, or if this is unknown. Where it is unknown, state any consequences about the inclusion of complexity classes that would follow from $A \leq_L B$. [9 marks]

Answer: We know that $REACH \leq_L CVP \leq_L SAT$ and $CVP \leq_L VAL$. Thus, by part (b) above, we also have $REACH \leq_L VAL$. All other reductions are unknown.

In particular, if $CVP \leq_L REACH$, this would imply that the complexity classes P and NL are equal. If $SAT \leq_L CVP$, it would imply $P = NP$. If $SAT \leq_L REACH$, it would imply $NL = P = NP$. If $VAL \leq_L CVP$, it would imply $P = NP = co-NP$. If $VAL \leq_L REACH$, it would imply $NL = P = NP$. If $SAT \leq_L VAL$ or $VAL \leq_L SAT$, then $NP = co-NP$.
